
CSCI2950P Final Project Report: Unsupervised Learning of Scene Categories

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Abstract

We present the Hierarchical Latent Dirichlet Allocation (hLDA) for organizing a collection of scene images into a hierarchy in unsupervised fashion. The bag of visual words model with spatial pyramids is adapted to the hLDA where the spatial generality is used as a prior of assignments of topics to visual words. Both depth and width of the tree are unbounded and inferred by training data using the Nested Chinese Restaurant Process (nCRP). We use a collapsed Gibbs sampler as the posterior inference algorithm that use the Metropolis-Hastings (MH) steps to obtain next values of hyperparameters. We experiment our algorithm on both toy data set and scene dataset. The results show that the unsupervised learning method can also achieve a decent classification rate.

1 Introduction

Classification is an essential task for scene understanding in computer vision researches. Traditional scene classification algorithms are limited for several reasons. At first, it assumes that each image belongs to only one scene category. In addition, all the categories need to be in a flat structure so that they are exclusive to each other. Thirdly, the number of categories is usually predefined and fixed to a constant number. All these assumptions do not hold for image set in real life. An typical scene images can belong to several categories in a semantic meaning from coarse to fine. For example, an image shows a beach in Florida should at first be categorized to the category “outdoor”. Then it may be analyzed more and more carefully by asking “is this a man made scene or natural one?” or “is it urban scene or rural scene?” Therefore, there is increasing demands that the space of categories should be better organized. On the other hand, as computer vision community becomes more data-driven, there are huge datasets with millions of images under construction. As a result, it is hard to predict what is the exact number of scene categories at a given time point. An algorithm that infer this number from data will help to eliminate the problem.

Although it is still controversial that whether a tree is the correct structure to classify images, there are already some previous works on constructing hierarchical structure for image collections. Supervised visual taxonomies were studied in several vision literatures [5][10][6]. In particular, Fei-Fei et al.[5] tries to construct an image hierarchy using crowdsourcing technique. However, it is known problem that human labeler can either make mistakes or be biased based on their unique visual experience. Massive labeling tasks will require a lot of human worker and therefore could be potentially financially inefficient.

On the other hand, Bayesian Nonparametric model has becoming popular for its capability of making inference based on data. In Natural Language Processing studies, the hierarchical Latent Dirichlet Allocation (hLDA) model has been invented to cluster documents based on a tree of topics [4]. Structure of the tree can be inferred from training data using the Nested Chinese Restaurant Pro-

cess (nCRP). The results shows that the model can put topics with coarse semantic meaning onto higher level of the tree which will be shared by multiple groups of documents. The success of the application in NLP domain make us wonder if we can use the same model in the scene clustering task.

In this project, we apply the hLDA model with nCRP prior to scene clustering problem. We constructed our own image dataset by downloading 1500 images (15 categories each has 100 images) from SUN database [11]. For each category, we use 50 images to construct the tree, and then use a label assignment algorithm to assign names of category to 15 of the tree nodes. Finally, we predict the labels of test images by using the KNN algorithm and assigning them to the node of their closest neighbors.

There are two similar work existing. Sivic et al. [9] uses the hLDA model to build an object class hierarchy. Our work is very similar with theirs but done on a scene dataset. Moreover, we used the learned structure to predict labels for heldout dataset and compare the result with the traditional supervised pipeline, while they didn't. Bart et al. [2] invented the TAX model to do visual taxonomies, which can be seen as an extension of original hLDA model but with one more layer of abstraction. The concept of category is introduced into the TAX model, which can be seen as multinoulli distributions of topics where number of state need to be fixed. A node on tree is tied with a category rather than a topic. They claimed that without introducing category, the model cannot be fitted into visual data well. However, in both work of Sivic et al. [9] and our work we found that by considering the spatial information into the model, images can still be correctly partitioned without any modification of the original hLDA model.

The report is organized as follows. We begin with a review of the necessary background of the hLDA model, nCRP as well as the MCMC method that used to do posterior inference in Section 2. In Section 3, we describe how we adapt the model with bag of visual words model and spatial pyramids, which has been proven to be state of the art representation of scene images. We present how the hierarchy is assessed using ground-truth labels and how it can be used to do predictions on heldout image data in Section 4. Our experiment on both toy dataset and an image dataset constructed from original SUN database are described in Section 5. Finally, we present the problem that still need to be addressed in the future in Section 6.

2 Statistical Model

2.1 The hierarchical Latent Dirichlet Allocation model

In the hLDA model [4], each document is assigned into one path c from root to the leaf of the tree. Because each node is associated with one topic, all words of the documents are comes from the topics in the path. That is to say, if the tree has a fixed length of L , then the document is a superposition of L topics with specific mixture weights. Assuming that the structure of tree T is already known, The words in a single document will be sampled using the generative process below: (1) For each document, pick a path c from the tree; (2) Sample a vector of L length θ from a Dirichlet distribution with parameter α , i.e. $p(\theta|\alpha)$, then construct a multinoulli distribution parameterized by θ ; (3) For each words in the document, sample the topic assignment z based on the multinoulli distribution; (4) Sample the word from another multinoulli distribution associated with the specific topic z , which controlled by another Dirichlet distribution $p(\beta|\eta)$.

The left image of Figure 1 shows the graphical model of the hLDA model. It is almost the same as the graphical model of the LDA model except that there is one more arrow pointing to w for the path assignment c , which comes from the tree T . Also notice that so far we assume that the structure of the tree T is known so there is no more parameters that controls T . However in practice, it is often difficult to specify a suitable structure of the tree in advance. Blei et al. [4] developed a hLDA model, which automatically infers the structure by placing a Nested Chinese Restaurant Process.

2.2 The nested Chinese Restaurant Process

The nCRP is a natural extension of original Chinese Restaurant Process in which each level can be seen as an independent CRP. For each level of the tree, a CRP specifies a distribution on partitions of documents into nodes in that level. The distribution determines the probability that how a new

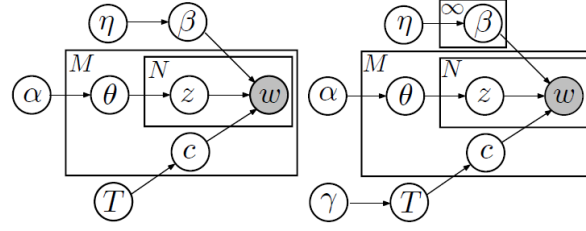


Figure 1: Left: The graphical model of the hLDA model. Right: The graphical model of the hLDA model with the nCRP as prior which has hyperparameter γ . M is the number of document and N is the number of words in the documents. Image credit: [9]

customer will be seated based on Eq.1

$$\begin{aligned}
 p(\text{occupied table } i | \text{previous customers}) &= \frac{n_i}{\gamma + n - 1} \\
 p(\text{next unoccupied table} | \text{previous customers}) &= \frac{\gamma}{\gamma + n - 1}
 \end{aligned} \tag{1}$$

where n_i is the number of customer currently sitting at table i , and γ is the real valued parameter which controls how often, relative to the number of customers in the restaurant, a customer will choose to sit at a new table. While in the nCRP model, the culinary metaphor works a little different as follows. The customer at first enters the root Chinese restaurant and select a table to sit based on the rules of CRP described in Eq. 1. On the table, there is a card on which there is a recommendation of another Chinese restaurant. At the second time, the customer will try the restaurant recommended at the first time. Another CRP is then used to select table for the customer. The process repeat on and on and restaurant recommended at level i will never be those appears in level 1 to $i-1$. The nCRP is controlled by hyperparameter γ and the graphical model of the hLDA with nCRP as prior of tree structure is shown in the right image of Figure 1. Notice that the depth of the tree is unbounded so the nCRP can repeat forever. However, in our implementation, we need to fix the depth to a constant L . This will not weaken our model as long as we set the value to be sufficiently large. Figure 2 illustrates a three level random tree generated by the process.

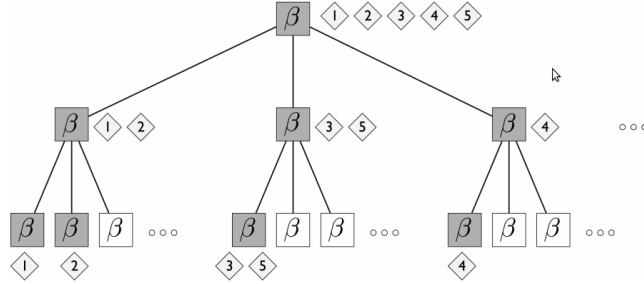


Figure 2: Visualization of a nCRP with three levels. Each box represents a restaurant, each diamond represents a customer. Image credit : Blei et al. [4]

2.3 Posterior Inference

A collapsed Gibbs sampler is used to perform the posterior inference for the hierarchical LDA model where β and θ in Figure 1 are integrated out. Conceptually, the process can be divided into three parts. At first, given the current state of the CRP, the topic z for each words in each documents are sampled following the algorithm developed in [7]. Second, given the values of the LDA hidden variables, the path assignment c is resampled based on the CRP. Finally, hyperparameters including γ, α and η will be updated in a Metropolis-Hastings step. In the implementation, α is replaced by

m and π in the two-parameter variant of stick breaking representation. The conditional probability distribution in Eq. 2 is used to sample c and z in each iteration. A clear explanation can be found in [3] which we will not rephrase here.

$$\begin{aligned} p(z_{d,n}|z_{-(d,n)}, c, w, m, \pi, \eta) &\propto p(z_{d,n}|z_{(d,-n)}, m, \pi) p(w_{d,n}|z, c, w_{-(d,n)}, \eta) \\ p(c_d|w, c_{-d}, z, \eta, \gamma) &\propto p(c_d|c_{-d}, \gamma) p(w_d|c, w_{-d}, z, \eta) \end{aligned} \quad (2)$$

3 Spatial pyramid with bag of visual words 0.75

The bag of words model is also well adapted to vision problems after its first success in linguistics studies. In computer vision, each images can be seen as a documents. Although visual world is much more complex than languages, one can reduce the complexity to a fix number of visual words. To do this, sufficiently large image patches are firstly collected. These patchese are usually local descriptors like SIFT, or other gradient based representations. The collection of image pathces are then clustered into a fixed number of groups. Then center for each group can then be seen as visual words. For any given images, the patches generated can be quantized into visual words based on some distance metric function like SSD. The resulting hitogram will be exactly the same as the ones we can get for text data.

Another technique that is common in image representation is spatial pyramid [8]. One weakness of original bag of visual words model is that the spatial information is completed discarded. However, to understand scenes in the images, it is often important to know the layout of the gradients. Spatial pyramid is added into the model to tackle this problem. Figure 3 illustrates the basic idea of the algorithm. The images are divided into several bins, in each bins, a local histogram is built based on the patches that fall into the grid. All the histograms are finally concatenated together and weighted in a proper way so that both local and global information are encoded in the representation.

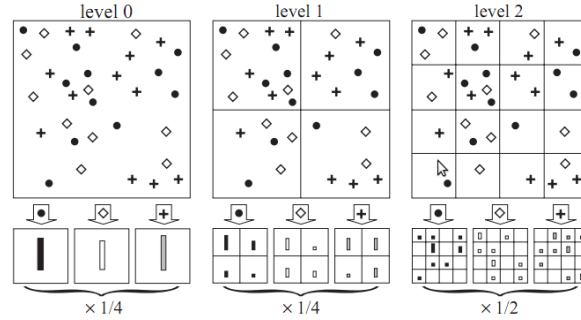


Figure 3: Example of a three-level pyramid. The image has three feature types, indicated by circles, diamonds, and crosses. At the top, the image are subdivided at three different levels of resolution. Next, for each level of resolution, the features that fall in each spatial bin are counted. Image credit: Lazebnik et al. [8]

4 Assessment of hierarchy using classification

To use the tree in image classification, we must find a way of matching the hierarchical partition with labels. We use the 'classification overlap score' defined in [9]. An scene category i and node t in the tree's overlapping can be metric by equation $\rho(i, t) = \frac{GT_i \cap N_t}{GT_i \cup N_t}$, where GT_i is the ground truth set of images of category i and N_t is the set of images which are assigned to a path passing through tree node t . Notice that the score ranges between 0 and 1 while higher score indicating better 'overlap' between the category i and node t . To obtain the overall assessment of the tree, we take the node with maximum overlap for each category and get a final score by averaging over the sum of all maximized class score: $\rho = 1/N_c \sum_i \max_t \rho(i, t)$, where N_c is the total number of ground truth categories in the dataset. The category assignment problem here is actually very difficult to solve when the number of categories increases. An combinatorial optimization need to be used to find out the best solution because each node in the tree can only be associated with one category. In the other

words, they have a one-to-one mapping relationship. However, this could not be true in real life, a category that is coarse enough may be used to describe two or more subtrees. In the future, we can devise a better algorithm for node-to-category mapping but by now we just use the existing one.

5 Experiments

In this section, we describe our experiments with the hLDA and nCRP model on two datasets. We use the code available online for Blei et al. [4]. In their implementation, the gibbs sampler will try 100 initial states with random assignments of path and topics for each documents. We used the same strategy for testing on toy data. However, some modifications have been made for our testing on image dataset, so that instead of try random initialization, we assign each visual words to a certain level based on its spatial generality. The detail will be described later in the paper. The inference algorithm is written in C. The image processing part, including computation of SIFT descriptor, construction of visual words as well as the implementation of spatial pyramids [8] is done in Matlab. We use the VLFeat package [1] to generate dense-SIFT descriptors for images.

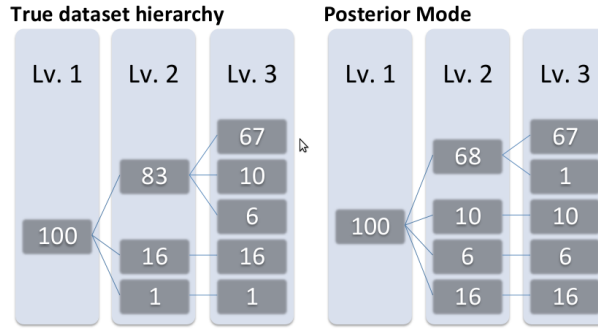


Figure 4: Estimated mode of the posterior hierarchy from hand-crafted data.

5.1 Experiment I: Toy dataset

We at first test our algorithm on a simulated data set, which contains 100 documents of 250 words each. The vocabulary size is fixed to 100. The structure of the tree is determined by the nCRP with γ set to 1 and depth number fixed to 3. The weights of topics on a path is generated by the two parameter version of stick breaking construction [4] with $gem_mean = 0.5$, $gem_scale = 100$. All topics are controlled by a Dirichlet distribution with parameter $\eta = 0.005$, which means that our topics are sparse. In this setting, each documents contains about 10-30 unique words. The left hand image in Figure 4 illustrate the dataset we have generated. It reveals the property of CRP that “the big gets bigger”.

After the dataset is generated, we use the same parameter settings to run our Gibbs sampler for 10 times, with maximum iteration number equals to 5000. The MCMC algorithm converges typically after 2000-3000 iterations. On the right hand in Figure 4, the estimated mode of the posterior distribution across the hierarchy and allocation is illustrated. Different from the result shown in Blei et al. [4], we cannot recover the right structure of the tree, although the all documents are correctly partitioned in the bottom level. The explanation of this phenomenon in the original nested CRP paper is that the accuracy of tree structure is depend on the size of dataset and how identifiable the topics are. While I find another reason could be that three level tree could be too shallow so that the truncation of the stick breaking happens very soon. As a results, words comes from the bottom level topics so frequently so that the partition in second level becomes a more difficult task. This breaks the expectation that more words should come from a coarse level of topics. We can expect this problem will not happen in real data set because topics in image data are more identifiable so there shouldn’t be problem for each visual words to choose the right level from a path.

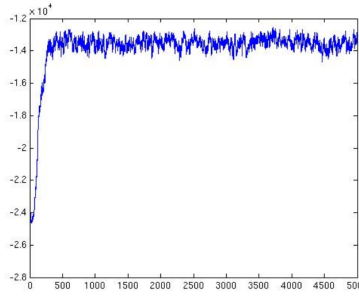


Figure 5: The complete log likelihood of Eq. 3 for the first 5000 iteration of the Gibbs sampler run on 60 scene images of 3 categories.

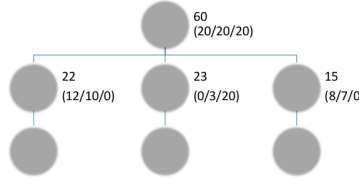


Figure 6: Estimated mode of the posterior hierarchy from 60 scene images of 3 categories. The numbers indicate images that allocated to the specific node. The numbers in the parenthesis indicate images in each individual category that is allocated to that node.

5.2 Experiment II: Image dataset of 15 scene categories

We then construct a dataset from images in SUN database [11]. We use 15 categories of images including indoor scenes such as office, kitchen, living room and bedroom, as well as outdoor scenes like street, highway, coast and mountains, etc. Each of the category contains 100 images. We discarded the color information and used only the gradient based SIFT descriptor.

Two visual vocabularies are then constructed as follows. For each of the images in our database, dense-SIFT descriptors are computed on scales 4 and 8. Around 1000 descriptors are generated per image from which we randomly sample 15% of them into a descriptor pool. After the sampling step, we send all descriptors in the pool into a k-mean cluster algorithm. The algorithm will return the means of each cluster which are taken as our visual vocabulary. K here equals to 10 and 100 respectively. After the visual vocabularies are constructed, we build the spatial pyramid representations for each image. We set the pyramid level equals to 3 for vocabulary with 100 words and 1 for that with 10 words. Therefore, the resulting spatial-SIFT vocabulary has word size equals to $10(1 \times 1) + 100(1 \times 1) + 400(2 \times 2) + 1600(4 \times 4) = 2110$. The dense-SIFT descriptors are then quantized into this vocabulary and a histogram is generated for each image. We realize that by using spatial pyramid, one descriptor will be counted for several times in the histogram, which means the number of word in images is increased. In the later step, we found that the running time of each iteration of our Gibbs sampler will be dramatically increased if words number becomes large. Therefore, we multiply the histogram vector by a universal scalar k here and do a rounding to the word counts. This keeps the number of visual words that appears in an image always around 200 -300. We found this approximation is important when we want to fit more data into the model. Another trick we have done here is that we discard the patches that have rare gradient information. To do this we put a threshold on the sum of SIFT descriptor, and all those less than the threshold are drawn out.

The image corpus file is then feed into our Gibbs sampler written in C. To accelerate the convergence of MCMC algorithm, assignments z of visual words to levels of the tree are initialized according to the generality of both appearance and spatial layout. Visual words that comes from the size 10 vocabulary are assigned to the root level node, words from the size 100 vocabulary are assigned initially to level 2, words from the second and third layer of the spatial pyramids, which contains 400

and 1600 words, are assigned to level 3 and level 4 respectively. The assignments can change during the model fitting so we don't need to worry if the heuristics is not accurate.

$$L^{(t)} = \log p(c_{1:D}^{(t)}, z_{1:D}^{(t)}, w_{1:D} | \gamma, \eta, m, \pi) \quad (3)$$

Two experiments have been done on this dataset. As the number of categories and images used, the tree becomes more and more semantically ambiguous, so at first, we use only 3 categories of 20 images for each for testing and illustration. We choose “coast”, “highway” and “store” as the categories to be tested. The depth of tree is again fixed to 3. We tested with different values of γ , finding that it has a great impact on branching factor of the tree. While setting γ to a small value like 0.01 will help us gain the real structure for this 3 category test. The Metropolis-Hastings update of γ is not helpful in this task because in this scenario we do want to use the BNP model to simulate the classification problem in which we have already known the real number of category. The Gibbs sampler mixes at iteration number 164 in this test case, giving a classification rate equal to 60.7% using the assessment method we described in Section 4. Figure 5 shows plot the joint log probability (Eq. 3) with respect to the number of iterations. With the initial assignments based on the appearance and spatial generality, the algorithm quickly finds the high probability area in the space. Figure 6 shows the allocation of images in the tree. We found that the category “store” is less confused with “coast” and “highway”. From Figure 7, we can better understand the reason. The latter two categories have much more similar spatial layout. For example, they both have long and dominant lines in the images, they are both less cluttered than images in category of “store”, etc.



Figure 7: Images from category “coast”(left), “highway”(middle) and “store”(right).

In the second test, we apply all the 15 categories of images to the model. For each category, we selected half of them as train and the other half as test data. We set $\gamma = 1$ and tree depth $L=5$ in this case and finally get 364 nodes from the tree. We used the ground truth label and combinatorial algorithm to assign category labels to the nodes that has largest overlapping scores. Then the method described in Section 4.2 of [2] is used to predict category assignment for test images. For each of the test image, we compute its histogram intersection score with all training images [8]. Then seven nearest neighbors is selected and vote for the node assignment. If all these images are in different leaf node, the 8th nearest neighbor is taken into consideration. The process repeats until there is at least two images vote for the same node. The test image is assigned to that node and its category assignment is decided. The image on the right in Figure 8 shows the final confusion matrix of our test data, the average accuracy is 60.0% compared to the overall accuracy 81.8% generated by the pipeline described in [8], which is shown on the left. The performance of our algorithm is way off this state of the art algorithm that uses 1 v.s. all kernel SVM. This result is somewhat expected because in this hLDA model, each image can only get information on one path while they are lack of information from other subtrees. On the other hand, we should argue that it is unfair to compare classification rate between the two models because the unsupervised method is not designed to recover ground truth labels. Instead, it is used to express the real structure of data and rectify the mistakes that may exist in ground truth data. Therefore, a missed classification doesn't mean that

our model is doing incorrect prediction, but that the data cannot be fit into the 15 flat classes very well and need some more sophisticated organizing strategy.

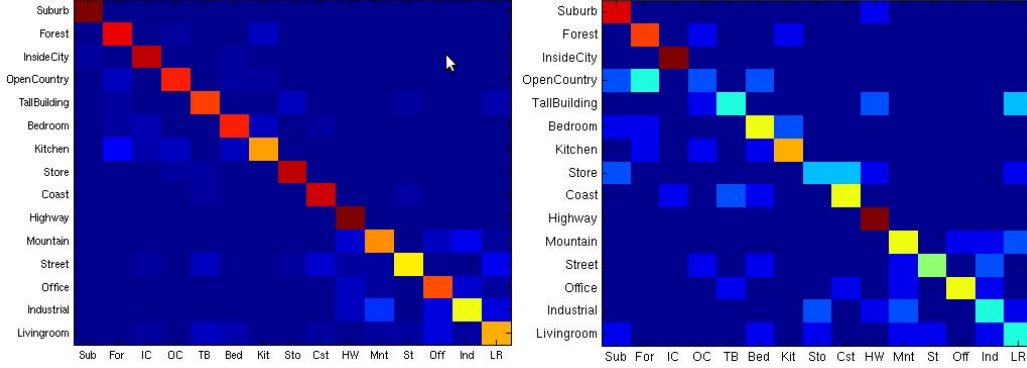


Figure 8: Right: Confusion matrix generated by pipeline described in [8] with average accuracy 81.8%. Left: Confusion matrix generated by our algorithm, with average accuracy 60.0%.

6 Discussion

Our experiments show that there are still problems worth of studying in the future. At first, compared to supervised methods, the MCMC algorithm runs very slow on image data. For 1500 images, it takes almost one day on a dual-core computer with 2GB memory. Considering more vision products begin to make use the huge dataset which often contains millions of images, this unsupervised learning technique still need to be improved to become a regular tool for vision community. Besides, our prediction methods, which is based on KNN algorithm, could also be very time consuming as the size of dataset increases. We have used the original code provided by Blei et al. [4] to do prediction but coping with the ground truth labels becomes very hard and tricky. On the other hand, the combinatorial algorithm introduced in section 4 will also become slow when the number of tree nodes grows. Generally it would be interesting to see if there are better algorithms to match labels with nodes on tree and propergate labels to all images in a partial labeled dataset. An automatic annotation algorithm will dramatically reduce the human intervention required on scene labeling tasks and therefore enable vision researchers to acquire massive image data with accurate labels more easily.

References

- [1] <http://www.vlfeat.org/>.
- [2] Evgeniy Bart, Ian Porteous, Pietro Perona, and Max Welling. Unsupervised learning of visual taxonomies. In *In CVPR*, pages 1–8, 2008.
- [3] David M. Blei, Thomas L. Griffiths, and Michael I. Jordan. The nested chinese restaurant process and bayesian nonparametric inference of topic hierarchies. *J. ACM*, 57:7:1–7:30, February 2010.
- [4] David M. Blei, Thomas L. Griffiths, Michael I. Jordan, and Joshua B. Tenenbaum. Hierarchical topic models and the nested chinese restaurant process. In *Advances in Neural Information Processing Systems*, page 2003. MIT Press, 2004.
- [5] Li Fei-fei. A bayesian hierarchical model for learning natural scene categories. In *In CVPR*, pages 524–531, 2005.
- [6] Sachin Gangaputra and Donald Geman. A design principle for coarse-to-fine classification. In *In Proceedings of CVPR*, pages 1877–1884, 2006.
- [7] Thomas L. Griffiths and Mark Steyvers. A probabilistic approach to semantic representation, 2002.
- [8] Svetlana Lazebnik and Cordelia Schmid. Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories. In *In CVPR*, pages 2169–2178, 2006.
- [9] Josef Sivic, Bryan C. Russell, Andrew Zisserman, Inria Ecole, and Normale Suprieure. Efros. unsupervised discovery of visual object class hierarchies. In *In Proc. CVPR*, 2008.

- [10] Gang Wang and Ye Zhang Li Fei-fei. Using dependent regions for object categorization in a generative framework. In *In CVPR*, pages 1597–1604. IEEE Computer Society, 2006.
- [11] Jianxiong Xiao, J. Hays, K.A. Ehinger, A. Oliva, and A. Torralba. Sun database: Large-scale scene recognition from abbey to zoo. In *Computer Vision and Pattern Recognition (CVPR), 2010 IEEE Conference on*, pages 3485 –3492, june 2010.